

Economic History II

The Guarani Jesuit missions and the persistence of human capital

FGV EPGE

The paper's epigraph

- *“When in 1768 the missions of Paraguay left the hands of the Jesuits, they had arrived at perhaps the highest degree of civilization to which it is possible to conduct a young people.”*
- Voltaire, quoted at the head of the article.
- An **anticlerical** writer, describing a Catholic order's settlements.

A human capital intervention that ended in 1767

- 1609 to 1767: the Society of Jesus runs missions, the **reductions**, among the Guarani, in today's Argentina, Brazil and Paraguay.
- Official aim: converting souls. In practice: reading, writing, arithmetic and crafts.
- 1767: the Spanish Crown expels the order. The missionaries never return.
- Valencia Caicedo's question: 250 years later, does the schooling still show up?
- Near former missions, educational attainment is 10% to 15% higher and income 10% higher today.

Three threads

- **Human capital transmission**: schooling passed from one generation to the next.
- **Economic persistence**: a shock that stopped in 1767 and is still measurable.
- **Culture**: occupations, work patterns and technology adoption.

THE MISSION: HUMAN CAPITAL TRANSMISSION, ECONOMIC PERSISTENCE, AND CULTURE IN SOUTH AMERICA*

FELIPE VALENCIA CAICEDO

This article examines the long-term consequences of a historical human capital intervention. The Jesuit order founded religious missions in 1609 among the Guaraní, in modern-day Argentina, Brazil, and Paraguay. Before their expulsion in 1767, missionaries instructed indigenous inhabitants in reading, writing, and various crafts. Using archival records, as well as data at the individual and municipal level, I show that in areas of former Jesuit presence—within the Guaraní area—educational attainment was higher and remains so (by 10%–15%) 250 years later. These educational differences have also translated into incomes that are 10% higher today. The identification of the positive effect of the Guaraní Jesuit missions emerges after comparing them with abandoned Jesuit missions and neighboring Franciscan Guaraní missions. The enduring effects observed are consistent with transmission mechanisms of structural transformation, occupational specialization, and technology adoption in agriculture. *JEL* Codes: I25, L16, N36, O15, O43, Q16, Z12.

The Guarani Jesuit reductions

A timeline

- 1534: the Society of Jesus founded in Paris by Ignatius of Loyola, *ad maiorem Dei gloriam*.
- 1580: first Guarani Franciscan mission. 1588: Jesuits reach Asunción.
- 1609: first Guarani Jesuit mission. Exploration period runs to 1659.
- 1750: Treaty of Madrid. 1767: expulsion.
- **158 years** of missionary intervention.

1534

1535

1580

1588

1609

1659

1750

1767

Jesuit
Order
founded
in Paris

Early
Spanish
Exploration
Routes

First
Guarani
Franciscan
Mission

Jesuits
arrive in
Asuncion

First
Guarani
Jesuit
Mission

End of
Jesuit
exploration
period

Treaty of
Madrid

Jesuits
expelled

The setting

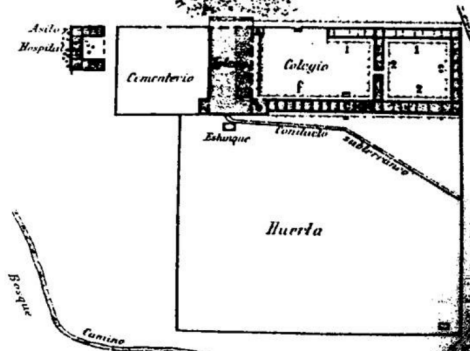
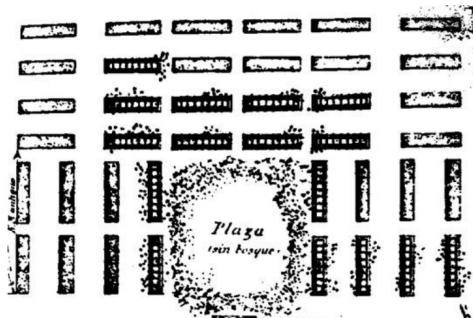
- One indigenous group across the whole area, the Guarani, the Tupis in Portuguese.
- At the priests' arrival, considered to be at a **Neolithic stage of development**, lacking iron weapons and tools.
- Semi-sedentary: manioc root and maize, by slash-and-burn agriculture.
- Uniform subtropical forest, no major mineral resources.
- Returns to human capital were presumably low: an unlikely place to invest in schooling.

The reductions

- **30 missions** in all: 15 in Argentina, 8 in Paraguay, 7 in Brazil (the *sete povos*).
- Planned settlements built around the church, the school and the workshops.
- Children taught to read, write and do basic arithmetic, boys and girls separately.
- Adults trained in masonry, wood carving and embroidery.
- At their peak more than 120,000 inhabitants, four times the population of Buenos Aires in 1779.

Inside a reduction

- Blueprint of San Ignacio Miní, drawn in 1899 by Juan Queirel.
- Central square, church, college, workshops, the garden (*huerta*), houses in regular blocks.
- A **small planned town**, not a village.



Where the missions were

- Misiones and Corrientes (Argentina), Itapúa and Misiones (Paraguay), Rio Grande do Sul (Brazil).
- Crosses mark the exact location of each Guarani Jesuit mission.
- **549 municipalities**, around 10 million inhabitants today.
- An area comparable in size to Uruguay or Ecuador.

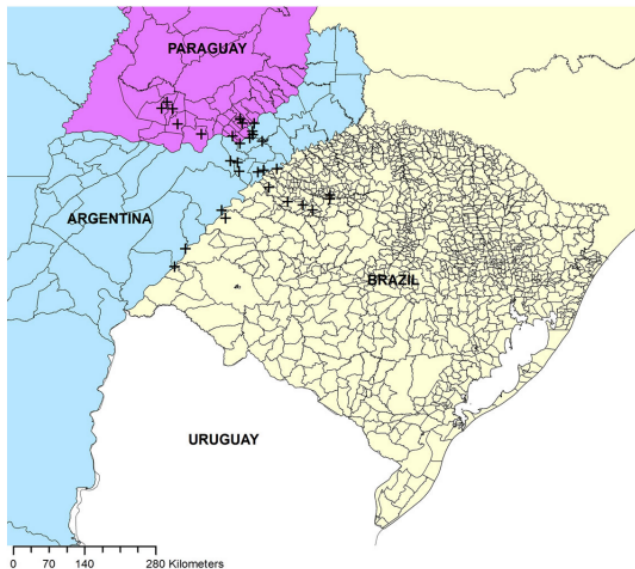


FIGURE I

Location of the Guaraní Jesuit Missions in Argentina, Brazil, and Paraguay

What is left standing

- São Miguel das Missões, Rio Grande do Sul.
- The scale of the stonework gives the measure of a reduction at its height.



The Spanish Jesuit mission of [São Miguel das Missões](#),
Brazil



A state within a state

- The missions stayed isolated from Spanish colonists and Spanish rule, and were only allowed to barter restrictively.
- The colonial capital itself was small: Asunción barely numbered **6,451** inhabitants in 1761.
- What drew the Guarani in was protection: inside a reduction they escaped enslavement and the forced labor of the *encomiendas*.

The missions among the Guaraní are often called collectively the [Río de la Plata](#) missions. The Jesuits attempted to create a "[state within a state](#)" in which the native peoples in the reductions, guided by the Jesuits, would remain autonomous and isolated from Spanish colonists and Spanish rule.^[4] A major factor attracting the natives to the reductions was the protection they afforded from enslavement and the forced labour of *[encomiendas](#)*.

Protection from the bandeiras

- Portuguese slave-hunting **bandeiras** out of São Paulo raided the reductions.
- The Jesuits produced guns and gunpowder inside the missions and raised Guarani militias.
- 1641: a Guarani militia defeats the Bandeirantes and their Tupi auxiliaries at Mbororé.
- Those raids also destroyed some early missions, which will matter for the identification.

The immediate need of the Guaraní in the 1640s was to protect themselves from slavers. The Jesuits began to arm them, producing guns and gunpowder in the missions.^[25] They also secured the Spanish Crown's permission, and some arms, to raise militias of Indians to defend the reductions against raids. The Bandeirantes followed the reductions into Spanish territory, but in 1641 the Guaraní militia defeated an army of 1,500 or more Bandeirante slavers and Tupi auxiliaries at the battle of Mbororé.^[4] The militias would eventually number as many as 4,000 troops and their cavalry was especially effective, wearing European-style uniforms and carrying bows and arrows as well as muskets.

Education at the center of the project

- Jesuits were at the **technological frontier** of the time: cartography, ethnography, linguistics, botany, mathematics, medicine.
- They introduced the printing press to Argentina, Brazil and Paraguay.
- They built an astronomical observatory at San Cosme y Damián, in modern-day Paraguay.
- Conversion was the stated aim; schooling and technical training were the daily practice.

It is hard to overemphasize the value of education for the Jesuit order. Jesuits were at the technological frontier of the time, and their cultural contributions to music and the arts are well known. [Furlong and Storni \(1994\)](#) stress the Jesuit achievements in cartography, ethnography, linguistics, botany, mathematics, and medicine. Jesuits introduced the printing press to Argentina, Brazil, and Paraguay and even established an astronomical observatory in San Cosme and Damián, in modern-day Paraguay. Jesuits played an important role in the historical development of tertiary education in Brazil ([Tobias 1972](#)). Even though the official aim of the Jesuit missions was to convert souls to Christianity, Jesuits taught children—boys and girls separately—how to read and write and do basic arithmetic ([Ganson 2003](#)). They also trained adults in masonry, wood carving, and embroidery ([Gálvez 1995](#)).¹³

The expulsion of the Jesuits

Pombal and the southern frontier

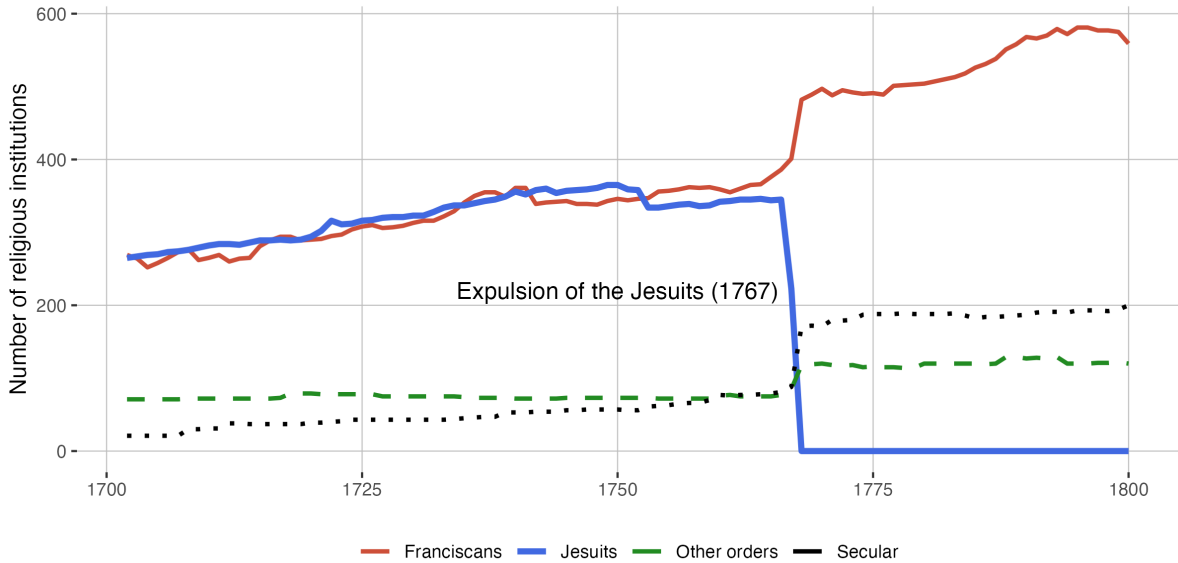
- Mid-eighteenth century: the Portuguese Crown, under the **Marquis of Pombal**, moves to assert control over the southern frontier.
- The Treaty of Madrid (1750) transferred the *sete povos* from Spanish to Portuguese rule.
- Guarani War, 1754 to 1756: a joint Portuguese-Spanish force defeats the mission militias.
- Pombal's charge, circulated across Europe: the missions were a state within a state.

Sepé Tiaraju

- Guarani leader from the *sete povos*, born around 1723, later a **popular hero** in Rio Grande do Sul.
- Led the refusal to leave the mission lands after the Treaty of Madrid: *“Esta terra tem dono”*.
- Killed on 7 February 1756. Three days later about 1,500 Guarani died at the battle of Caiboaté.

The suppression of the Society of Jesus

- 1759: Portugal confiscates Jesuit property and imprisons or exiles its members.
- France 1764, the Spanish Empire 1767, dissolution by Clement XIV in 1773.
- Charles III and Joseph I, counseled by Pombal, pressed the Pope; the governor of Buenos Aires, Bucarelli, carried the order out.
- By 1768 **no Jesuit missionaries remained** in the Guarani area. The order was restored in 1814, but never returned.



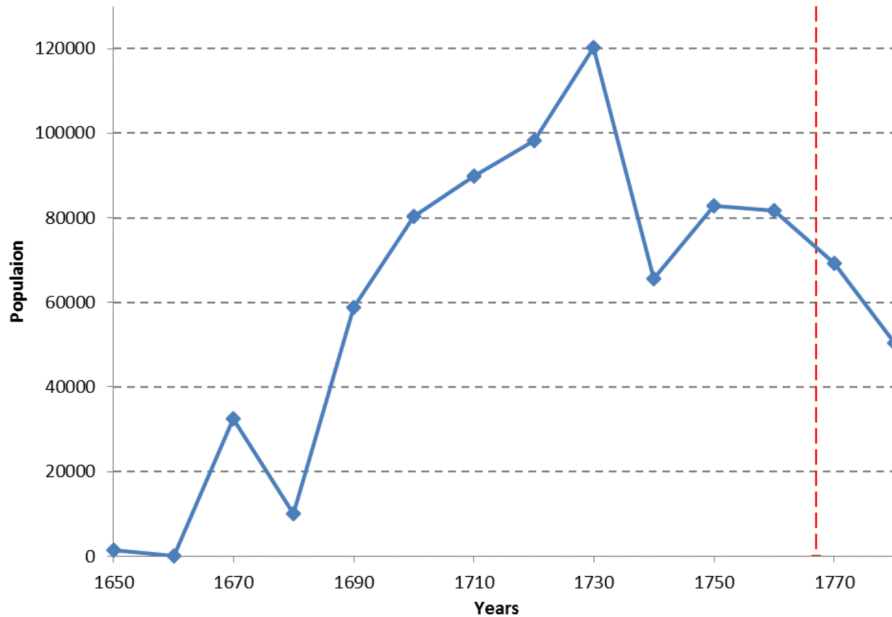
Where the title comes from

- The film **The Mission** (1986).
- The Guarani reductions and their destruction after the Treaty of Madrid.



An intervention cut off

- Mission population climbs for a century, past 120,000.
- The dashed line is **1767**.
- After the expulsion the missions emptied out, and the area saw outmigration.
- The treatment stops in 1767. The question is what survives it.



What is left, 250 years later?

How the missionary effect is measured

- Treatment: **distance** from the municipality centroid to the nearest Jesuit mission.
- Places near a former mission compared with places farther away, inside the same Guarani area.
- Geographic and weather controls: altitude, ruggedness, slope, temperature, rainfall, distance to the nearest river and coast.
- One indigenous group, three countries: national institutions and precolonial tribes are held aside.

The data

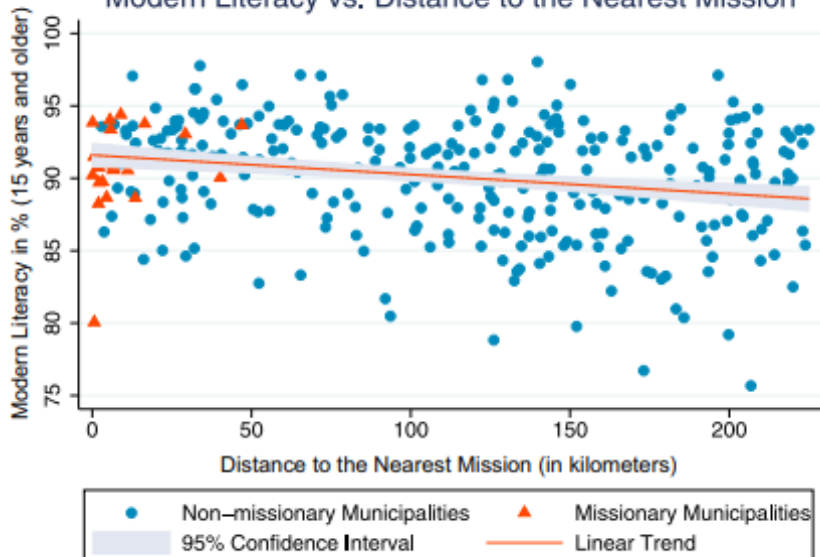
- Archives: the *Cartas Anuas* (annual letters) of the Jesuit fathers, the Archivo de Indias in Seville, the national archives of Argentina and Paraguay, the Roman Jesuit Archives.
- From them: location, year of foundation and population of every mission, geocoded.
- Modern censuses: Argentina 2001 and 2010, Brazil 2000 and 2010, Paraguay 2002 and 2012.
- Historical censuses: Argentina 1895, Brazil 1920, Paraguay 1950.
- Franciscan mission locations and populations, from the other order's records.

Literacy today falls with distance to a mission

- Each point is a municipality. Triangles had a mission, circles did not.
- Missionary municipalities cluster in the upper left corner, above 90% literacy.
- 100 km closer to a mission: illiteracy lower by at least 10%.
- In Brazil, median years of schooling rise by **0.67 years**, almost 15%.
- Comparable to the educational benefits attributed to Bolsa Familia.

(a)

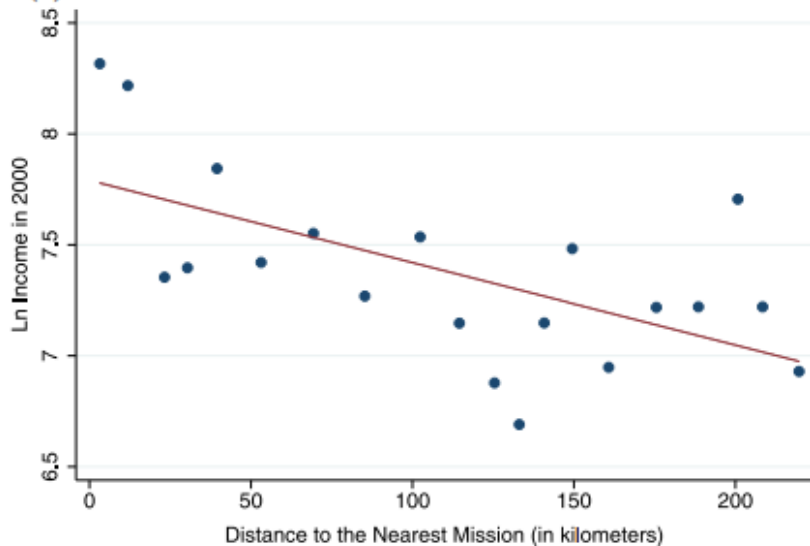
Modern Literacy vs. Distance to the Nearest Mission



Income follows

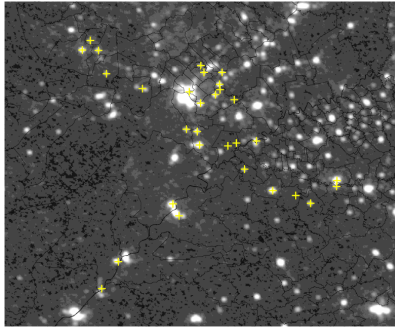
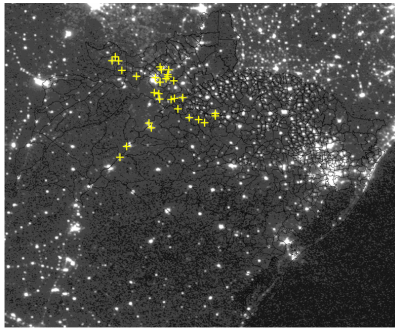
- Income per capita in 2000, Brazil and Paraguay, averaged within distance bins.
- 100 km closer: income higher by about **0.2 log points**, roughly 10% near former missions.
- Argentina and Paraguay, poverty instead of income: 100 km farther and the Unsatisfied Basic Needs index rises by about 10 points.
- The schooling difference shows up as an income difference.

(c) Modern Income vs. Distance to the Nearest Mission



The same pattern from space

- Nighttime satellite images of the Guarani missionary area, crosses marking the missions.
- The **lit points** sit where the reductions stood, in an otherwise peripheral region.



Tracking the shock through time

- Historical censuses: Argentina 1895, Brazil 1920, Paraguay 1950.
- The gap was **larger** then: in 1895 Argentina, 100 km closer meant illiteracy lower by 5%.
- Brazil in 1920 started from illiteracy of 35%, and the missionary gap was wide.
- Postwar the effect decays but survives, alongside general educational convergence in Latin America.
- In 1895 the gap is wider for women than men, and concentrated among Argentineans rather than foreigners: it travels through families, not through arrival.

Did the Jesuits pick better places?

Where the Jesuits actually settled

- The obvious worry: missions placed where prospects were already good.
- Jesuits were the **last major Catholic order** to arrive in the Americas, so they had last pick and ended up in peripheral areas.
- Places close to Jesuit missions have worse geographic and weather attributes, not better.
- Jesuit states were not denser in precolonial times.
- Two comparisons carry the argument: abandoned Jesuit missions, and Franciscan Guarani missions.

Abandoned missions

- Guayrá, Alto Paraná and Itatín: missionary nuclei founded in the exploratory period, 1609 to 1659, then abandoned.
- Sites the Jesuits chose with the same judgment, but which never received the **full treatment**.
- Abandoned after slave-hunting raids and the death of priests, at a time of low missionary recruitment in Europe: one or two Jesuits per mission.
- No consistent effect on education or income, and the Jesuit effect holds when they are controlled for.
- What mattered was the activity carried out, not the site chosen.

Jesuits and Franciscans, side by side

- Same indigenous group, same region, same religion, a different order.
- Franciscans were a **Mendicant** order: strict poverty vows, tending the sick and the poor, no comparable emphasis on schooling and technical training.
- Franciscans arrived first (1580) and settled further north, closer to existing population centers and on better land.
- Franciscan distance on its own: no effect on illiteracy today.
- With both orders in the same comparison, the Jesuit benefit holds and the Franciscan sign turns the other way.

TABLE VI
FRANCISCAN AND JESUIT MISSIONARY EFFECT ON MODERN EDUCATION

	Illiteracy						
	Argentina, Brazil, and Paraguay				Brazil	Argentina	Paraguay
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Franciscan mission distance	0.00579 (0.005) {0.006}	-0.00899 (0.008) {0.011}	-0.01200 (0.008) {0.009}	-0.0335*** (0.008) {0.012}	-0.0618*** (0.019) {0.019}	-0.0529* (0.029) {0.030}	-0.03500 (0.025) {0.026}
Jesuit mission distance			0.0131** (0.005) {0.005}	0.0183*** (0.005) {0.006}	0.0518*** (0.009) {0.010}	0.111*** (0.029) {0.032}	0.0208 (0.024) {0.029}
Geo controls	No	Yes	No	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	549	548	549	548	467	42	39
Within R^2	0.026	0.062	0.036	0.077	0.116	0.674	0.494
R^2	0.031	0.067	0.041	0.082	0.120	0.737	0.495

Notes. The table shows the coefficients for distance to the nearest Franciscan and Jesuit missions in kilometers (equation (1)). The dependent variable is illiteracy for people aged 15 years and older in 2000 in percentages for Argentina, Brazil, and Paraguay. Geographic controls include distance to the nearest coast, distance to the nearest river, altitude, ruggedness, temperature, area, rainfall, latitude, and longitude. Please refer to Section I of the Online Appendix for units and additional details of these variables. Estimation is by OLS with state fixed effects. Robust standard errors are in parentheses and Conley standard errors are in curly brackets *** $p < .01$, ** $p < .05$, * $p < .1$.

Why does the effect cross centuries?

Three mechanisms of transmission

- **Occupational specialization**: instruction and technical training moved people out of agriculture into a proto-artisan class.
- **Structural transformation**: less agriculture, more manufacturing and commerce.
- **Technology adoption**: more schooling means faster take-up of new techniques.
- Underneath all three, vertical transmission: educated parents, educated children, decade after decade.

Structural transformation

- Brazil 2010, Argentina 2001, Paraguay 2012, by sector of employment.
- Closer to a former mission: **less agriculture**, more manufacturing and more commerce.
- The same direction in all three countries, despite different national trajectories.

TABLE IX
MISSIONARY EFFECT ON STRUCTURAL TRANSFORMATION IN BRAZIL, ARGENTINA, AND PARAGUAY

	Brazil 2010 Employed in			Argentina 2001 Employed in			Paraguay 2012 Employed in		
	Agriculture (1)	Manufacturing (2)	Commerce (3)	Agriculture (4)	Manufacturing (5)	Commerce (6)	Agriculture (7)	Manufacturing (8)	Commerce (9)
Panel A: Without Geographic Controls									
Mission distance	0.0174 {0.027}	-0.0587*** {0.012}	0.00737 {0.008}	0.00543*** (0.002)	-0.00022 (0.001)	-0.00164*** (0.001)	0.0133** (0.006)	-0.00755 (0.005)	-0.0112** (0.005)
Observations	466	466	466	48,476	48,476	48,476	1,962	1,962	1,962
Within R^2	0.045	0.221	0.032	0.079	0.001	0.007	0.044	0.014	0.026
R^2	0.168	0.274	0.104	0.105	0.002	0.008	0.061	0.014	0.026
Panel B: With Geographic Controls									
Mission distance	0.151*** {0.048}	-0.0877*** {0.022}	-0.0300** {0.013}	0.0218*** (0.008)	-0.000228 (0.003)	-0.00908*** (0.003)	0.0147* (0.009)	-0.0189*** (0.007)	-0.0171*** (0.005)
Observations	426	426	426	48,476	48,476	48,476	1,928	1,928	1,928
Within R^2	0.135	0.287	0.190	0.148	0.013	0.014	0.100	0.050	0.054
R^2	0.301	0.357	0.272	0.167	0.014	0.014	0.117	0.051	0.054

Skill-intensive industries

- The ten industries ranked highest on **human capital intensity**: iron and steel, metals, chemicals, electric machinery, transport equipment, beverages.
- Closer to a former mission: more people working in practically all of them.
- The least skill-intensive sectors (leather, apparel, footwear, textiles, furniture, wooden products): no effect, or the opposite one.
- Skilled workers and the industries that need them sit in the same places.

TABLE X
MISSIONARY EFFECT ON SKILL-INTENSIVE INDUSTRIES IN BRAZIL

HCINT	Brazil 2010									
	Iron and steel	Tobacco products	Nonferrous metals	Fabricated metal products	Plastic products	Beverages industries	Transport equipment	Electric machinery	Industrial chemicals	Chemicals other
	11.425 (1)	11.509 (2)	11.547 (3)	11.577 (4)	11.678 (5)	11.967 (6)	12.346 (7)	12.357 (8)	12.704 (9)	13.031 (10)
Mission distance	-0.00255** (0.001)	-0.000104*** (0.000)	-0.0265*** (0.009)	-0.00230*** (0.001)	-0.000349 (0.003)	-0.0103*** (0.003)	-0.00552** (0.002)	-0.00685* (0.004)	-0.00367* (0.002)	-0.00203 (0.002)
Geo controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	174,964	174,964	174,964	174,964	174,964	174,964	174,964	174,964	174,964	174,964
Within R^2	0.025	0.007	0.046	0.072	0.077	0.026	0.123	0.051	0.013	0.044
R^2	0.025	0.009	0.046	0.075	0.084	0.034	0.124	0.055	0.024	0.046

Genetically engineered soy

- Brazil, 1996 to 2006: **GE soy seeds** become available.
- Closer to a former mission: a higher share of land harvested with GE soy. Farther away: more non-GE varieties.
- Agricultural productivity is higher near the missions as well.
- GE soy is labor saving, so those places also shift workers out of farming and into manufacturing.

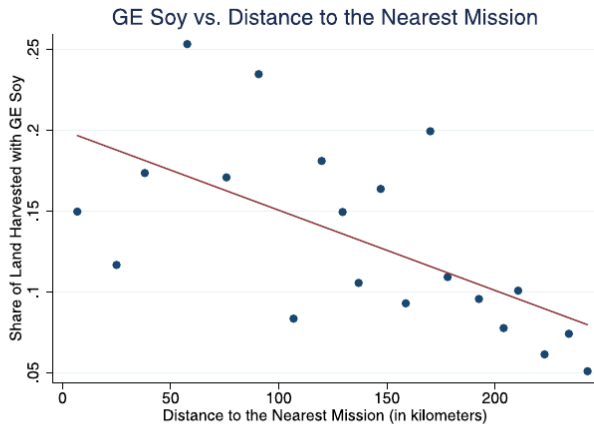


FIGURE V

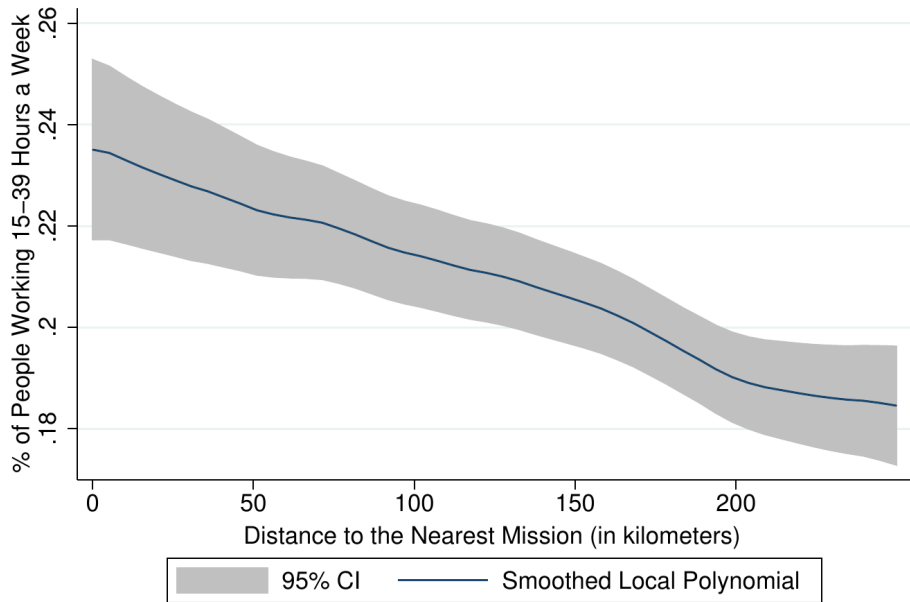
Binscatter of Technology Adoption in Agriculture on Missionary Distance:
Genetically Engineered Soy in Brazil

Unconditional binned scatter plot of share of land in farms harvested with GE soy in Brazil in 2006 versus distance to the nearest Jesuit mission in kilometers. The sample is restricted to a distance of 225 km. Color version available online.

Culture and work

- Closer to a former mission: a higher share of people working 15 to 39 hours a week, and higher labor force participation.
- Weber placed the Catholic orders in the same lineage as the Protestant ethic: *“The gradual rationalization of asceticism into an exclusively disciplinary method reached its apex in the Jesuit order.”*
- The naturalist d’Orbigny, visiting the former mission of Loreto in 1827, found people still living “following the old missionary customs”.

Hours Worked vs. Distance to the Nearest Mission



What the effect is not

- Not **urbanization**: places near Jesuit missions are less dense today, and their states were not denser before the conquest.
- Not migration: the schooling gap is found among residents, not among people who moved in.
- Not tourism: no effect on touristic activity in Brazil, though museum and monument visits are higher in Paraguay.
- Complementary investments do appear: denser roads, and a higher health index in Brazil.

The same question elsewhere

Table 6: Missionaries

Study	Setting	Method	Summary
<i>Missionaries, human capital, and health: Africa</i>			
Gallego and Woodberry (2010)	Africa	Empirical	Regions where Protestant missionaries dominated have higher literacy; role of competition
Nunn (2014)	Africa	Empirical	In areas with Protestant missionaries, education effect more pronounced for women
Wantchekon, Novta, and Klačnja (2015)	Benin	Empirical	Village-level externalities: descendants of the uneducated in villages with schools do better than those in control villages
Doyle, Meier zu Selhausen, and Weisdorf (2019)	Uganda	Empirical	Christian African converts adopted healthier sexual behavior already during colonial period
Cagé and Rueda (2020)	Africa	Empirical	Christian missionaries influenced sexual beliefs and behaviors that affected risk of HIV contagion
Wietzke (2015)	Madagascar	Empirical	Colonial institutions had stronger effect on local economic outcomes than missionary schools
<i>Missionaries, human capital, and health: Asia</i>			
Ma (2020)	China	Empirical	Jesuits missionaries stimulated Confucian literati to study science
Bai and Kung (2015)	China	Empirical	Protestant missionaries associated with schools, hospitals, and better economic development
Calvi, Mantovanelli, and Hoehn-Velasco (2019)	India	Empirical	Protestant missionaries associated with higher literacy
Calvi and Mantovanelli (2018)	India	Empirical	Current health outcomes better the closer a village to historical Protestant medical missions
Castelló-Climent, Chaudhary, and Mukhopadhyay (2018)	India	Empirical	Catholic missions associated with higher secondary school completion
<i>Missionaries, human capital, and health: Latin America</i>			
Waldinger (2017)	Mexico	Empirical	Areas with historical presence of Catholic mendicant orders have higher literacy and education today
Valencia Caicedo (2019a)	S. America	Empirical	Jesuit missions in South America promoted education; effects persist 250 years later
<i>Missionaries and other outcomes</i>			
Nunn (2010)	Africa	Empirical	Christian missionaries had lasting impact on religious beliefs
Woodberry (2012)	Worldwide	Empirical	Protestant missionaries influenced rise and spread of stable democracy
Cagé and Rueda (2016)	Africa	Empirical	In regions close to Protestant missions, proximity to printing press associated with higher newspaper readership, trust, education, and political participation
<i>Missionary expansion</i>			
Meier zu Selhausen (2019)	Africa	Empirical	Africanization of missions relevant to understand reach of initially European missionaries
Jedwab, Meier zu Selhausen, and Moradi (2019)	Africa	Empirical	Smaller effect size of missionaries when considering local determinants of missionary expansion

Benin: the first colonial schools

- Wantchekon, Klašnja and Novta (2015): the first regional schools in colonial Benin, early twentieth century.
- Student-level records: who enrolled in the first cohorts, and where those families are now.
- School locations and student cohorts were picked with **very little information**, so treated and untreated look alike at the start.
- Two comparisons: treated and untreated in the same village, and villages where no school was set up.

Benin: externalities

- The first generation of students and their descendants: higher living standards, less likely to be farmers, more politically active.
- **Village-level externalities**: descendants of the uneducated in school villages do better than those in control villages.
- Extended family: nephews and nieces benefit from their uncle's education, a “family tax” on the educated.
- Valencia Caicedo can follow places. Benin can follow named pupils.

EDUCATION AND HUMAN CAPITAL EXTERNALITIES: EVIDENCE FROM COLONIAL BENIN*

LEONARD WANTCHEKON
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NATALIJA NOVTA

Using a unique data set on students from the first regional schools in colonial Benin, we investigate the effect of education on living standards, occupation, and political participation. Since both school locations and student cohorts were selected with very little information, treatment and control groups are balanced on observables. We can therefore estimate the effect of education by comparing the treated to the untreated living in the same village, as well as those living in villages where no schools were set up. We find a significant positive treatment effect of education for the first generation of students, as well as their descendants: they have higher living standards, are less likely to be farmers, and are more likely to be politically active. We find large village-level externalities—descendants of the uneducated in villages with schools do better than those in control villages. We also find extended family externalities—nephews and nieces directly benefit from their uncle's education—and show that this represents a “family tax,” as educated uncles transfer resources to the extended family. *JEL* Codes: N37, O15, J27.

An educated child is like a lantern in your house at night.

—Eloi Gainsi, farmer and religious instructor,
Zagnanado (Benin)

Illustrative Example: Zagnanado. One missionary school was built in Zagnanado in the Agonlin region—a province of the Kingdom of Dahomey—in 1895 by Catholic missionaries from the Société des Missions Africaines in Lyon. The missionaries had increased access to the interior of Benin following the fall of the Kingdom of Dahomey in late 1894. Yet as of 1895, the French had not yet instituted a formal colonial administration in the area.¹¹

The pamphlet “Centenaire de l’Arrivée des Missionnaires au Pays Agonlin,” based on diaries and reports of founders of the Catholic school of Zagnanado, provides interesting details of the process leading to the creation of the school (see Map A.10 in Online Appendix A).

Figure A.1: Historical Map of Benin, Four School Sites Selection

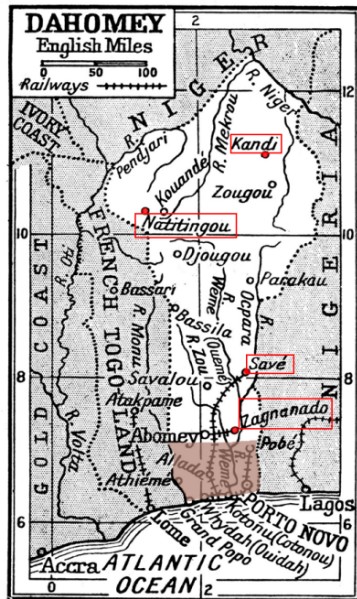
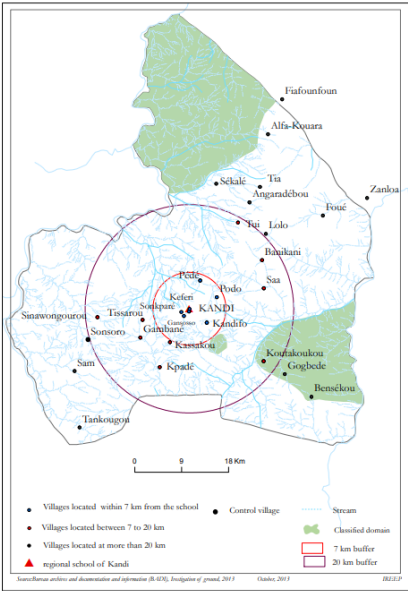


Figure A.2: Map of Kandi



Mexico: which order, and for whom

- Waldinger (2017): **1,145** mission locations in colonial Mexico, Mendicant and Jesuit.
- Mendicants (Franciscans, Dominicans, Augustinians): committed to poverty, educating the native population in rural mission areas.
- The Jesuits in Mexico did the reverse: educational effort aimed at the colony's elite in the city centers.
- So the order's label is held constant and the target of the teaching varies.

Mexico: the results

- Mendicant mission areas: higher literacy today, and higher attainment at primary, secondary and post-secondary levels.
- Jesuit mission areas: no such effect on the population's education.
- The share of **Catholics** is higher wherever any Catholic mission stood. The education gap is not.
- The mirror image of the Guarani case: among the Guarani it was the Jesuits who taught the population.

The long-run effects of missionary orders in Mexico[☆]



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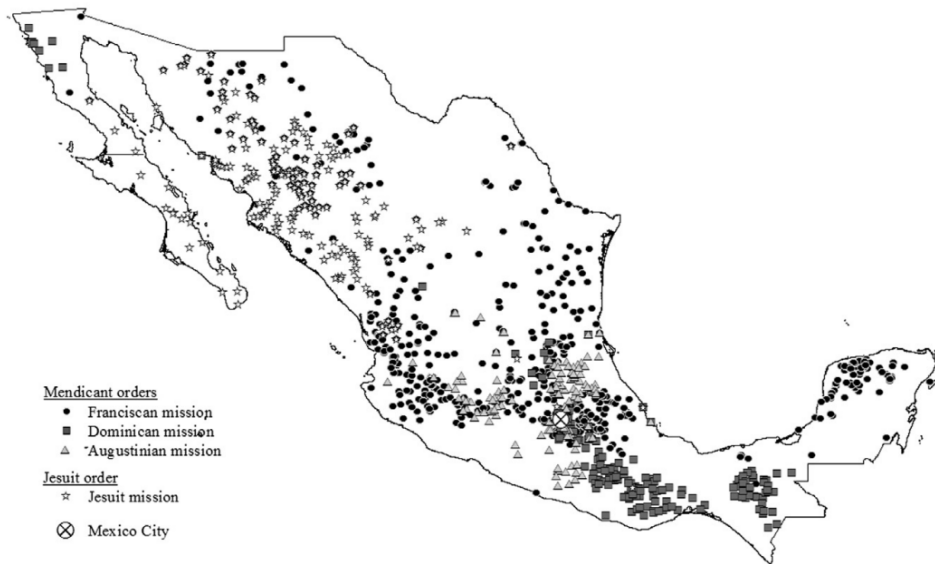
Missionary orders

Colonial Mexico

ABSTRACT

This paper examines the long-run effects of different Catholic missionary orders in colonial Mexico on educational outcomes and Catholicism. The main missionary orders in colonial Mexico were all Catholic, but they belonged to different monastic traditions and adhered to different values. Mendicant orders were committed to poverty and sought to reduce social inequality in colonial Mexico by educating the native population. The Jesuit order, by contrast, focused educational efforts on the colony's elite in the city centers, rather than on the native population in rural mission areas. Using a newly constructed data set of the locations of 1,145 missions in colonial Mexico, I test whether long-run development outcomes differ among areas that had Mendicant missions, Jesuit missions, or no missions. Results indicate that areas with historical Mendicant missions have higher present-day literacy rates, and higher rates of educational attainment at primary, secondary and post-secondary levels than regions without a mission. Results show that the share of Catholics is higher in regions where Catholic missions of any kind were a historical present. Additional results suggest that missionaries may have affected long-term development by impacting people's access to and valuation of education.

Catholic Mission Stations in Colonial Mexico, 1524 - 1810



Africa: missions with and without a press

- Cagé and Rueda (2016): Protestant missions in nineteenth-century sub-Saharan Africa, geocoded from a 1903 map.
- Protestant missionaries were the first to import the **printing press** and to let the local population use it.
- Some mission stations had a press, others did not.
- Each station's printing, educational and health investments are recorded separately.

Africa: the effect of the technology

- Within regions close to missions, proximity to a press goes with higher newspaper readership, trust, education and political participation.
- The comparison group is **other missions**, so the question becomes the technology rather than the missionary.

The Long-Term Effects of the Printing Press in sub-Saharan Africa[†]

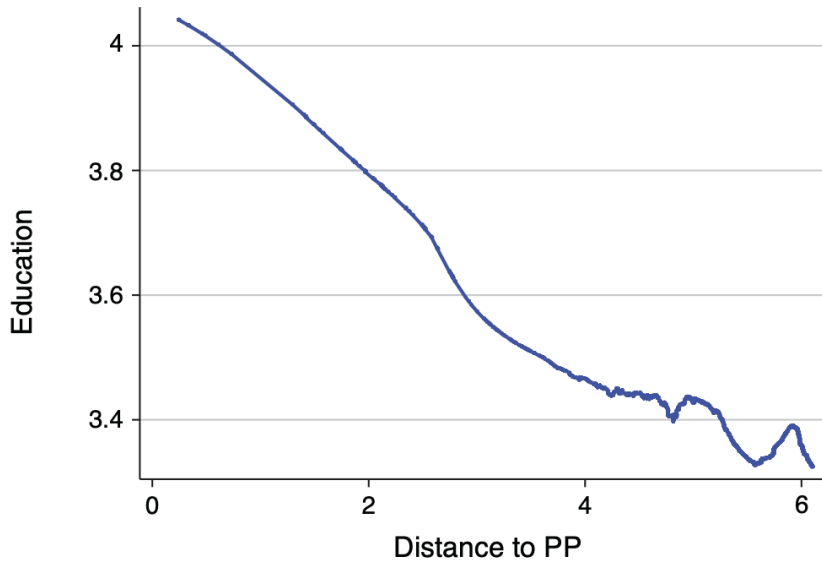
By JULIA CAGÉ AND VALERIA RUEDA*

This article investigates the long-term consequences of the printing press in the nineteenth century sub-Saharan Africa on social capital nowadays. Protestant missionaries were the first to import the printing press and to allow the indigenous population to use it. We build a new geocoded dataset locating Protestant missions in 1903. This dataset includes, for each mission station, the geographic location and its characteristics, as well as the printing-, educational-, and health-related investments undertaken by the mission. We show that, within regions close to missions, proximity to a printing press is associated with higher newspaper readership, trust, education, and political participation. (JEL L82, N37, N77, N97, O17, O43, Z13)

TABLE 2—DETERMINANTS OF THE LOCATION OF THE PRINTING PRESS
(comparing missions with and without the printing press)

	No printing (1)	Printing (2)	Difference (3)
<i>Geographical characteristics</i>			
Number of growing days (percent)	53.123	60.049	−6.926 (4.920)
Suitability for rainfed crops	4.992	4.750	0.242 (0.356)
Average precipitation per day (over year)	3.413	3.791	−0.378 (1.208)
Distance to 2,000 city (100 km)	2.702	2.655	0.047 (0.374)
Distance to the coast (100 km)	2.310	2.747	−0.437 (0.534)
Malaria ecology	5.187	10.295	−5.108 (1.509)
<i>Historical characteristics</i>			
Slave exports, per capita	0.074	0.042	0.031 (0.107)
Railway contact	0.237	0.107	0.129 (0.081)
Explorer contact	0.143	0.286	−0.143 (0.069)
Initial population density/100	11.803	13.760	−1.958 (8.008)
Distance to 1,400 city (100 km)	8.872	6.969	1.903 (0.932)
Distance to 1,800 city (100 km)	15.587	10.704	4.883 (1.890)
Observations	651	28	679

Panel C. Education



Conclusion

- 250 years after the expulsion: educational attainment 10% to 15% higher and income 10% higher near the former missions.
- Abandoned Jesuit sites show nothing, Franciscan missions show nothing. What is left is the **teaching**.
- It persists through occupation, sector and technology adoption, and not through cities, migration or tourism.
- Larger in the historical censuses, decaying but visible after the war.
- Same literature as AJR and Nunn, different channel: schooling rather than extractive institutions.

The author's own caveat

- *“Far from advocating historical determinism, a thorough understanding of these historical forces, and their implied constraints, offers the opportunity to make modern economic policies more targeted and effective.”*